

The Moon

The Moon is Earth's only natural satellite. It is one of the biggest moons in the Solar System compared to the planet that it orbits, and its gravity has a significant effect on Earth. However, it does not have enough gravity to cling to an atmosphere, and this makes it a very different world from ours.



▲ WANING MOON

Over the two weeks following a full moon, the area lit by the Sun gets smaller every night, dwindling to a bright crescent and eventually an almost invisible "new moon."

ORBIT AND PHASES

The Moon takes just over 27 days to orbit Earth. It spins on its axis at the same rate, so the same side always faces us. During its orbit, the angle between the Earth, Moon, and Sun changes, altering how much of its surface seems to be illuminated when viewed from Earth. Once a month, the whole face is lit up, but the bright area gradually shrinks (wanes) to nothing, and then grows bigger (waxes) again.

AIRLESS WORLD

The Moon has only one-sixth of Earth's gravity, which is not enough to hold on to an atmosphere. This means that there is no air, no weather, and nothing to protect the surface from solar radiation by day or to retain heat at night. Each Moon day lasts for 13 Earth days, so almost two weeks of scorching heat are followed by the same period of extreme deep freeze. Any water is either instantly boiled away or frozen in deep, permanently dark craters. This is why there is no life on the Moon.

FAST FACTS

- **Average distance from Earth** 237,034 miles (381,470 km)
- **Size (diameter)** 2,160 miles (3,476 km)
- **Mass compared to Earth** 0.0123
- **Average surface temperature** -9.4°F (-23°C)
- **Maximum surface temperature** 225°F (107°C)
- **Minimum surface temperature** -243°F (-153°C)
- **Day length** 13 Earth days
- **Orbit period** 27.3 Earth days

